

Sharif University of Technology
Department of Mechanical Engineering

Modeling of an RPP Robot in the Simulink Environment

Robotics Course – Homework 2

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1 Problem Statement

At the outset, we are asked to simulate the RPP robot model in the MATLAB Simulink environment and to display the relevant reference coordinate frames on the given figure.

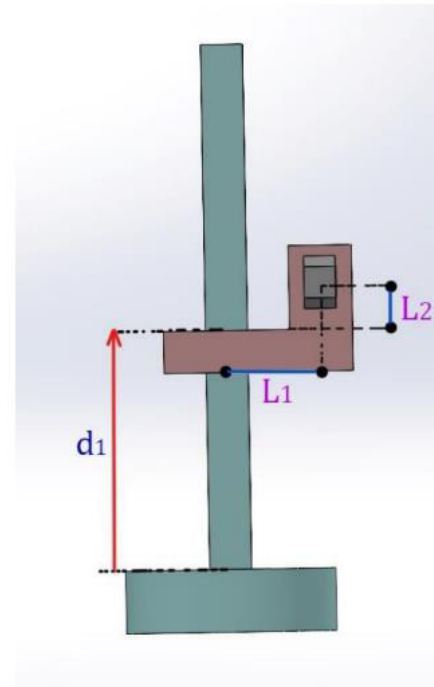
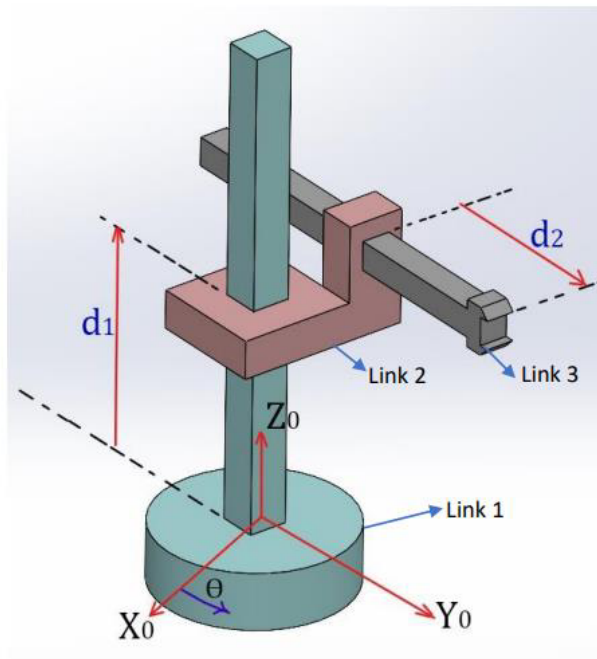
1.1 Question 1

In the first part of the question, we are asked to use the IC (Initial Condition) block to apply the given values as initial conditions to each of the joints, and then, based on this, to find the end-effector coordinates in each case.

1.2 Question 2

In the second part, we must use the Denavit–Hartenberg (DH) method to attach the corresponding coordinate frames to each link and fill in the associated DH table. Then, using the transformations obtained, for each of the six cases given in the first part, we find the homogeneous transformation from frame 3 to frame 0, and thereby obtain the end-effector coordinates using this method as well, comparing the results with those of the first part.

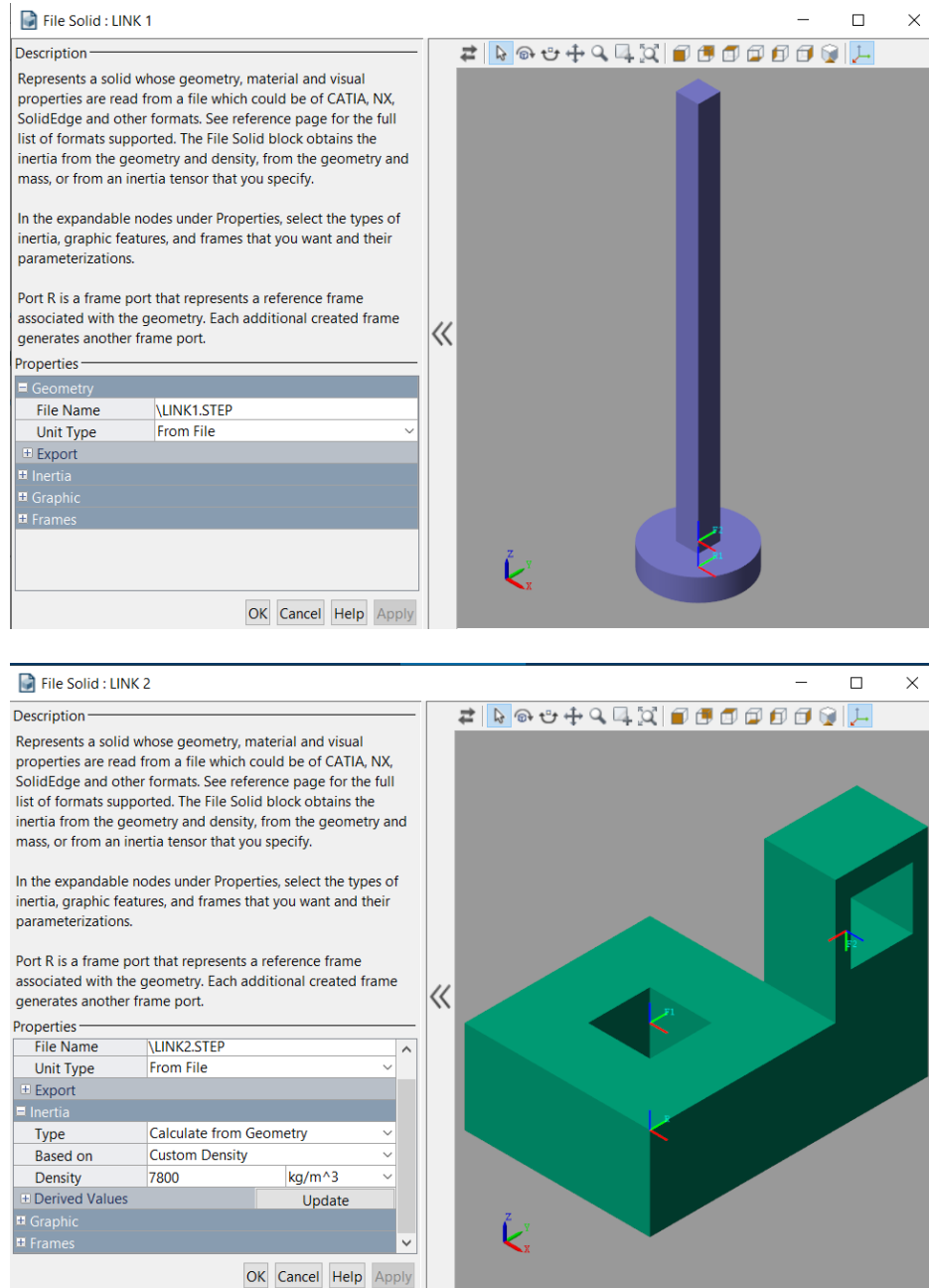
θ (deg)	-90	30	150	-18	56	280
d_1 (mm)	150	120	30	100	30	220
d_2 (mm)	100	80	0	0	170	110



2 Answers to the Questions

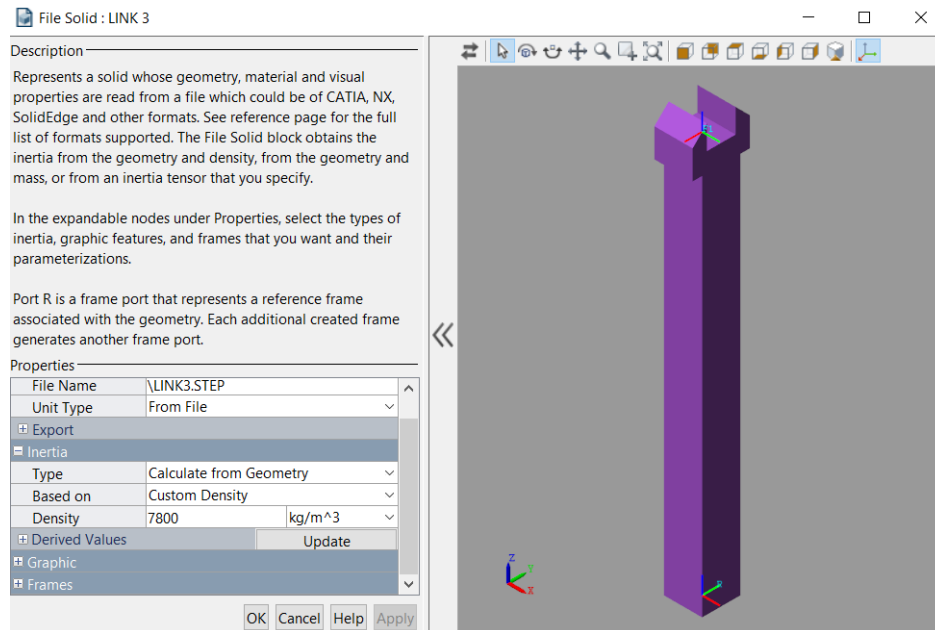
2.1 Question 1

We first draw the links in SolidWorks, and then open them in the Simulink environment using the **Solid File** block. We treat all parts as steel and manually enter its density in MATLAB:



Frames appropriate to the use of each link have been mounted on it as needed. Then, using one Revolute joint and two Prismatic joints, we assemble the given mechanism.

Instead of using the IC block, we use the **State Targets** and **Specify Position Target** sections in the settings of each joint, and in the initial state we set the conditions as shown below, in order to have an overall view of the mechanism:



Revolute Joint : Revolute Joint

Description

Represents a revolute joint acting between two frames. This joint has one rotational degree of freedom represented by one revolute primitive. The joint constrains the origins of the two frames to be coincident and the z-axes of the base and follower frames to be coincident, while the follower x-axis and y-axis can rotate around the z-axis.

In the expandable nodes under Properties, specify the state, actuation method, sensing capabilities, and internal mechanics of the primitives of this joint. After you apply these settings, the block displays the corresponding physical signal ports.

Ports B and F are frame ports that represent the base and follower frames, respectively. The joint direction is defined by motion of the follower frame relative to the base frame.

Properties

Z Revolute Primitive (Rz)

State Targets

Specify Position Targ...
☒

Priority
Low (approximate)

Value
56
deg

Specify Velocity Targ...
☐

Internal Mechanics

Limits

Actuation

Sensing

Mode Configuration

OK
Cancel
Help
Apply

Prismatic Joint : Prismatic Joint

Description

Represents a prismatic joint between two frames. This joint has one translational degree of freedom represented by one prismatic primitive. The joint constrains the follower origin to translate along the base z-axis, while the base and follower axes remain aligned.

In the expandable nodes under Properties, specify the state, actuation method, sensing capabilities, and internal mechanics of the primitives of this joint. After you apply these settings, the block displays the corresponding physical signal ports.

Ports B and F are frame ports that represent the base and follower frames, respectively. The joint direction is defined by motion of the follower frame relative to the base frame.

Properties

Z Prismatic Primitive (Pz)

State Targets

Specify Position Targ...
☒

Priority
Low (approximate)

Value
50
mm

Specify Velocity Targ...
☐

Internal Mechanics

Limits

Actuation

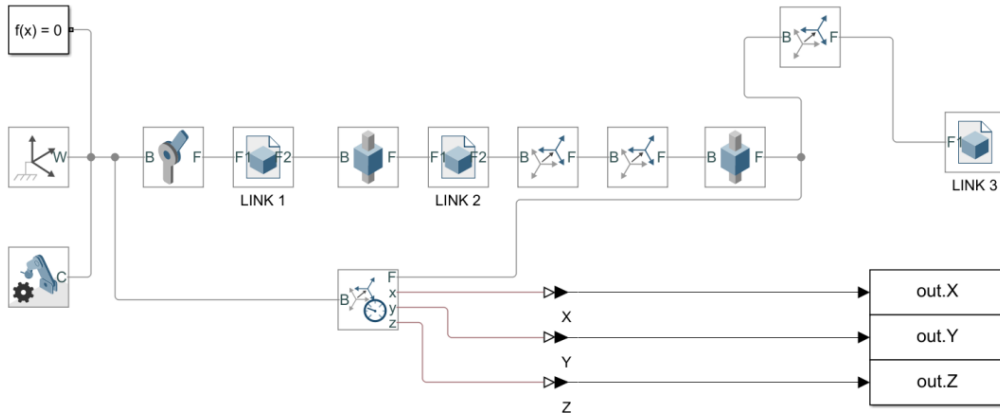
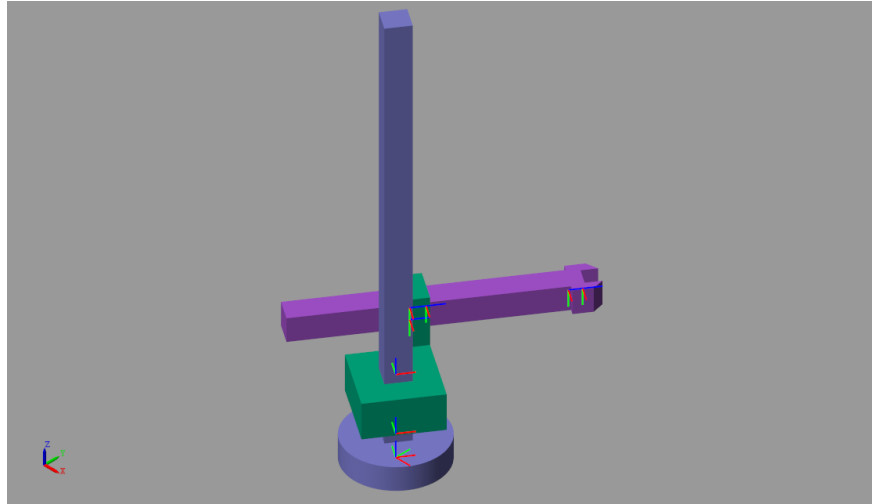
Sensing

Mode Configuration

OK
Cancel
Help
Apply

5

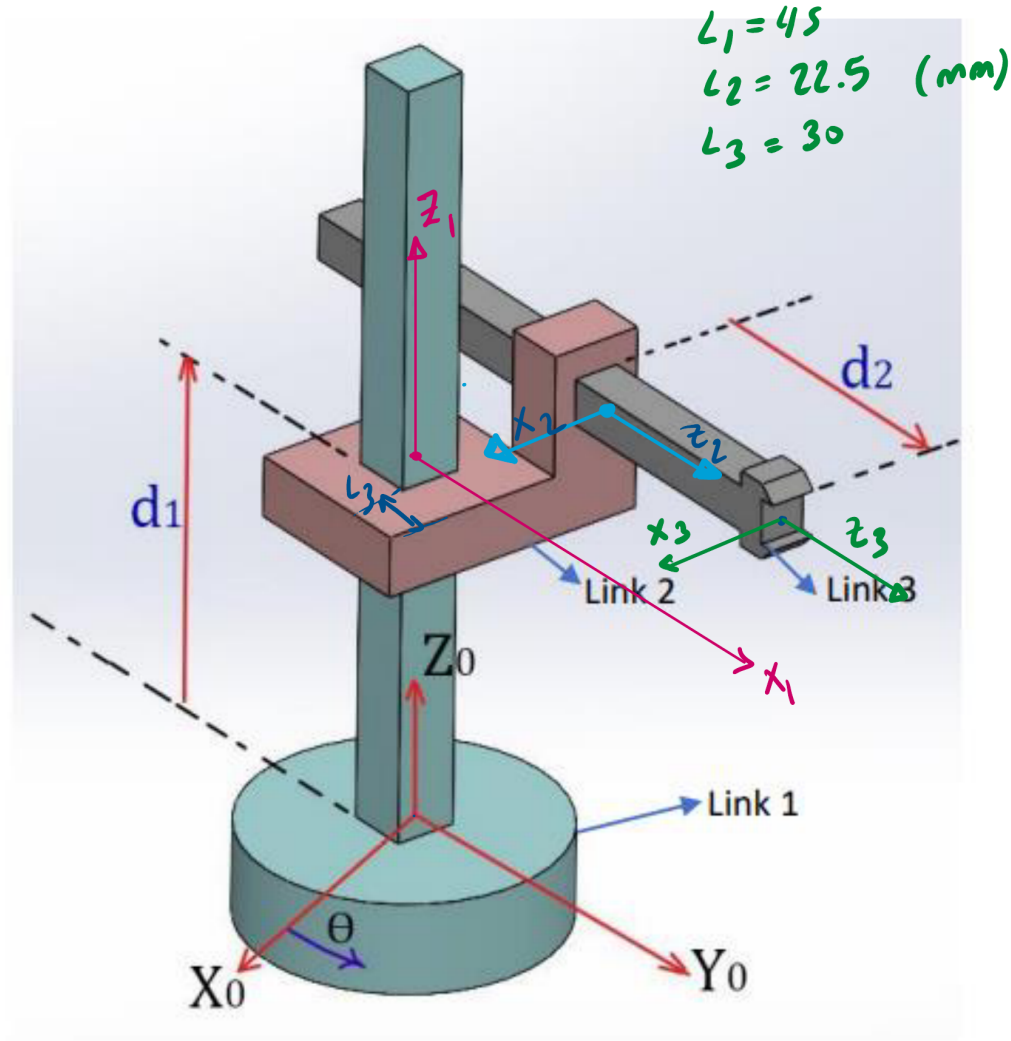
Now, for the six stated cases, using **State Targets** in the settings of each joint, we set the given values and record the end-effector position with respect to frame 0 for each case. To read the end-effector position we use the **Transform Sensor** block:



Mode	1	2	3	4	5	6
θ_1	-90	30	150	-18	56	280
d_1	150	120	30	100	30	220
d_2	100	80	0	0	170	110
x (mm)	45	72.8	-48.5	42.4	74.5	68.6
y (mm)	-130	94.0	-24.0	33.5	191.0	-130.1
z (mm)	192.5	162.5	72.5	142.5	72.5	262.5

2.2 Question 2

We apply the Denavit–Hartenberg method to the given mechanism as follows:



The Denavit–Hartenberg table will be as follows:

	θ	d	l	α
1	θ_1	d_1	0	0
2	-90	L_2	$-L_1$	-90
3	0	$d_2 + L_3$	0	0

$$H_3^0 = R_{z,\theta_1} T_{z,d_1} R_{z,-90} T_{z,L_2} T_{x,L_1} R_{x,-90} T_{z,d_2+L_3} \quad (L_3 = 30 \text{ mm}) \quad (1)$$

$$P_0^0 = H_3^0 P_0^3 \quad (2)$$

$$P_0^3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (3)$$

In order to compute the end-effector position for each of the 6 cases using the Denavit–Hartenberg method, we write the following MATLAB script and run it. The output of this script gives us a table in which all 6 cases above are placed.

```
theta1 = [-90,30,150,-18,56,280];
d1 = [150,120,30,100,30,220];
d2 = [100,80,0,0,170,110];
L1 = 45;
L2 = 22.5;
L3 = 30;
P00 = ones(4,6);

for i = 1:6
    H03 = Rot('Z', theta1(i))*Trans('Z',d1(i))*Rot('Z',-90)*Trans('Z',L2)*
        Trans('X', -L1)*Rot('X',-90)*Trans('Z',d2(i)+L3);
    P00(:,i) = H03*[0; 0; 0; 1];
end

P00 = P00(1:3,:);

table = [theta1;d1;d2;P00(1,:);P00(2,:);P00(3,:)]
```

Script output:

```
table =
-90.0000    30.0000   150.0000   -18.0000    56.0000   280.0000
150.0000   120.0000    30.0000   100.0000    30.0000   220.0000
100.0000    80.0000         0         0   170.0000   110.0000
 45.0000    72.7628   -48.4808    42.4375    74.5319    68.6271
-130.0000    93.9711   -23.9711    33.5270   190.9712   -130.0589
172.5000   142.5000    52.5000   122.5000    52.5000   242.5000
```

The above table, arranged as follows, will look like this:

Mode	1	2	3	4	5	6
θ_1	-90	30	150	-18	56	280
d_1	150	120	30	100	30	220
d_2	100	80	0	0	170	110
x (mm)	45	72.7628	-48.4808	42.4375	74.5319	68.6271
y (mm)	-130	93.9711	-23.9711	33.5270	190.9712	-130.0589
z (mm)	172.5	.1425	52.5	122.5	52.5	242.5

It is observed that all the numbers in the tables from Question 1 and Question 2 are equal to one another, with the exception of the z row. The 20 mm discrepancy between the z -row values in the two tables arises from the fact that, in the Simulink simulation, our reference frame is located at the bottom of the cylindrical base section of Link 1, and the thickness of this section is 20 mm. For this reason, the z -coordinate results obtained from the simulation are 20 mm greater than the results obtained with the Denavit–Hartenberg method.

*The codes and related files are included in the appendix.